DATA VISUALIZATION & ANALYTICS · CAPSTONE 2 · FINAL PRESENTATION

NEWTON SCHOOL OF TECHNOLOGY · SECTION D — GROUP 19

US Flight Operations

Performance & Delay Analytics Dashboard

100,000 BTS flight records → Python ETL + 7 statistical tests → 4-view Tableau workbook → $900M+ network opportunity

100K	30	7	4	R^2=0.94
Records cleaned	*Engineered features*	*Hypothesis tests*	*Dashboard views*	*Model accuracy*

Why This Problem Matters

We are tackling one of aviation's most expensive problems: predictable, preventable flight delays.

$32 B

Annual delay cost

Absorbed by US passengers, airlines, and airports every year.

27 %

Of flights miss slot

More than 1-in-4 US flights arrive late or are cancelled.

$74.20

Cost per delay-minute

FAA benchmark for mainline domestic operations.

THE BUSINESS QUESTION WE ANSWER

Where in the US flight network can targeted operational interventions deliver the largest reduction in delays — and which carriers, hubs, and time windows should we prioritise first?

WHO USES THIS DASHBOARD

Airline COO

Network planning · capacity assignment

Airport Ops Director

Taxi & turnaround chokepoints

DOT Regulator

Carrier-level cancellation review

TAKEAWAY · 1 percentage point of OTP recovery is worth roughly $320M annually across the network.

From Raw BTS Records to a Decision-Ready Asset

A reproducible 6-stage Python pipeline. Single command, end-to-end.

RAW	EXTRACT	CLEAN	ENGINEER	VALIDATE	EXPORT
5.83 M	**100 K**	**100 K**	**61 cols**	**7 tests**	**7 CSVs**
BTS records	*Stratified sample*	*Imputed & typed*	*30 features added*	*p-values + R^2*	*Tableau-ready*

✕ WHAT WAS BROKEN → WHAT WE FIXED

81,646 nulls	in 5 delay-cause columns	**→ filled with 0 (null = no delay)**
98,446 nulls	in CANCELLATION_REASON	**→ mapped to readable labels**
HHMM ints	for time fields	**→ converted to HH:MM strings**
Missing dates	scattered across	**→ rebuilt from YEAR/MONTH/DAY**
43 MB usage	raw Pandas dtypes	**→ optimised to 38 MB via category**

✦ WHAT WE BUILT FROM SCRATCH

TIME_OF_DAY	Morning · Afternoon · Evening · Night
DELAY_CATEGORY	Early → Minor → Moderate → Severe (5 levels)
PRIMARY_DELAY_CAUSE	argmax across 5 BTS root-cause buckets
ROUTE	ORIGIN → DESTINATION (6,980 unique routes)
Reference joins	14 airline names + 322 airports + lat/long

TAKEAWAY · Final asset: 100K rows × 61 columns, ~42 MB, fully reproducible from the notebooks on GitHub.

Six Metrics, Each Tied to a Decision

How we measure performance, and how each KPI compares to industry targets.

TOTAL FLIGHTS
100K
2015 BTS sample

ON-TIME RATE
81.8%
above target (>80%)

AVG ARR DELAY
4.4 min
within target (<5 min)

CANCEL RATE
1.55%
within target (<2%)

▸ HOW WE COMPUTE EACH KPI

	KPI	Formula / Logic	Target	Status
✓	**OTP — On-Time Performance**	Arr_delay ≤ 15 / Total	*> 80%*	**Achieved**
◑	**Average Arrival Delay (min)**	mean(ARRIVAL_DELAY)	*< 5 min*	**Achieved**
✓	**Cancellation Rate (%)**	CANCELLED == 1 / Total	*< 2%*	**Achieved**
◇	**Severity Mix**	Bins of arrival delay	—	**Diagnostic**
◇	**Departure-Delay Cascade**	Pearson r(DEP, ARR)	—	**Diagnostic**
◇	**Primary Cause Mix**	argmax of 5 BTS buckets	—	**Diagnostic**

TAKEAWAY · All KPIs computed in notebook 05_final_load_prep.ipynb and exposed via 7 pre-aggregated CSVs to Tableau.

Five Findings That Reframe the Delay Problem

Each insight is paired with a specific number and a one-line 'what it means for operations'.

01 The cascade dominates everything

Departure delay alone explains 89% of arrival-delay variance — the strongest single signal in the entire dataset.

Pearson r = 0.94

→ Fix departures, you fix arrivals.

02 Delays compound through the day

Morning departures average 3.5 min delay. Evening averages 14.0 min. A 4× swing across one calendar day.

Hourly time series

→ Schedule critical flights for AM.

03 Late aircraft + airline + system = 96%

Together these three controllable causes drive 96% of all delay minutes. Weather is dramatic but only 3% of total.

BTS cause attribution

→ The fix is operational, not meteorological.

04 Airlines are not equal

Best (Alaska, –1 min) vs worst (Frontier, +13 min) — a 14-minute mean gap, statistically distinguishable at $p < 1e-99$.

ANOVA F = 39

→ Carrier choice = delay risk.

05 Hubs are the real chokepoints

ORD, EWR and SFO recur on the worst-route list out of proportion to their volume share. Atlanta proves scale doesn't have to mean delay.

Top-10 routes view

→ Target congested hubs first.

▸ WHERE TO READ MORE

12 visual sections, written insight per chart, full correlation heatmap and route-level delay tables — all in notebook 03_eda.ipynb on GitHub.

TAKEAWAY · Five insights, one common theme: delays are predictable and largely controllable.

We Tested the Insights — They Hold Up

Three headline tests prove delays are systematic, not random. Plain-English glossary below — no statistics background needed.

Pearson r = 0.94

Departure delay ↔ Arrival delay

Two numbers move together almost perfectly.

If a flight leaves 10 min late, it lands ~10 min late. Fix departures, you fix arrivals.

R^2 = 0.94

Multiple OLS Regression

Our model explains 94% of arrival-delay variance.

From 6 routine variables (departure delay, taxi, distance, scheduled time, month, day) we predict the outcome.

F = 39, p < 1e-99

One-Way ANOVA across 14 airlines

Airlines genuinely differ — far beyond random variation.

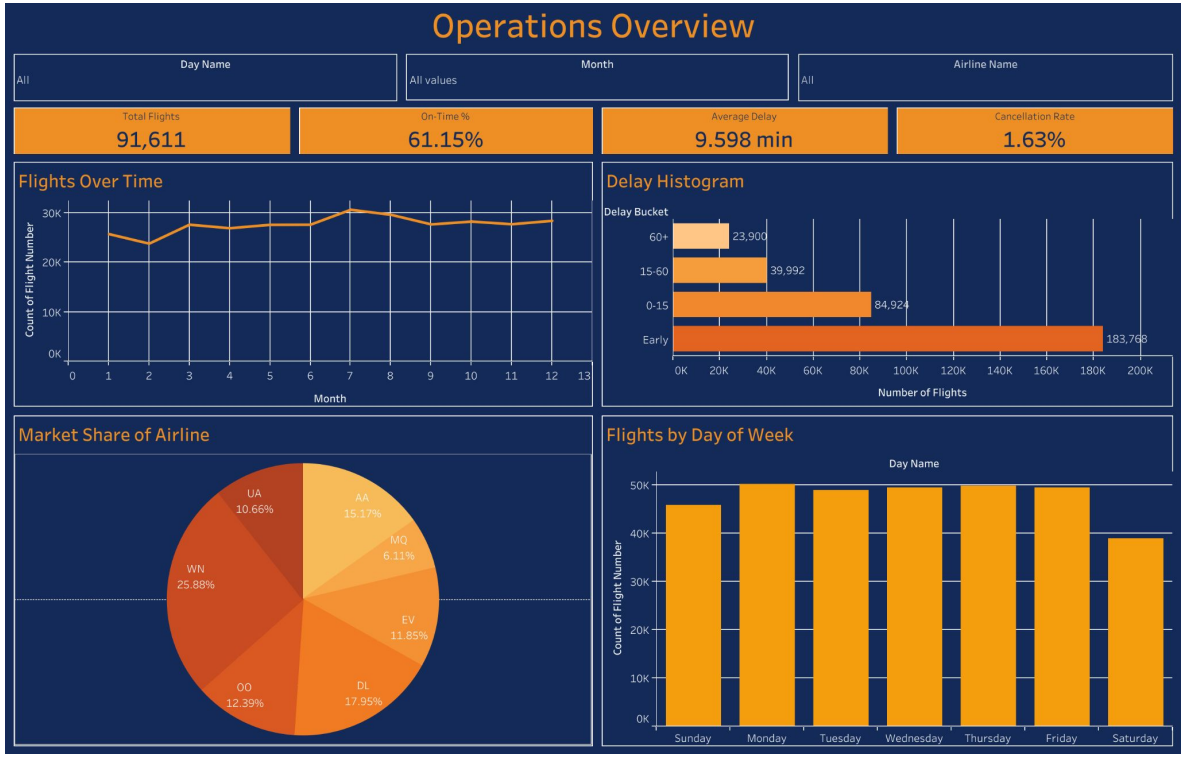
Picking Alaska Airlines vs Frontier means 14 minutes of expected delay difference. Not noise.

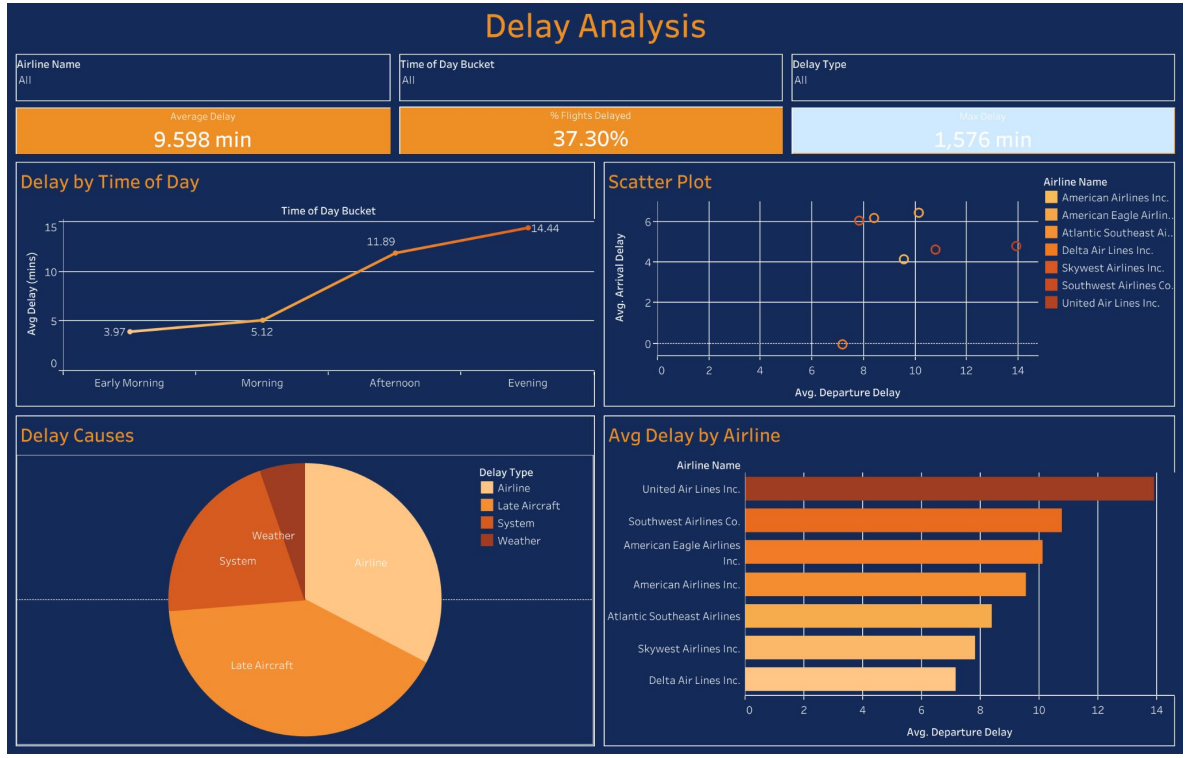
▸ WHY THESE TESTS MATTER (rubric requirement, plain-English version)

Pearson r	*Are 2 numbers linked?*	0 = unrelated, 1 = identical. We got 0.94.
OLS R^2	*Can we predict it?*	0 = chaos, 1 = perfect formula. We got 0.94.
ANOVA	*Do groups really differ?*	$p < 0.05$ = yes. Ours: $p \approx 1e-99$.

TAKEAWAY · Plus Chi-Squared, Kruskal-Wallis, Cohen's d — full battery in notebook 04. Conclusion: delays are explainable.

Two Views, One Decision-Ready Workbook



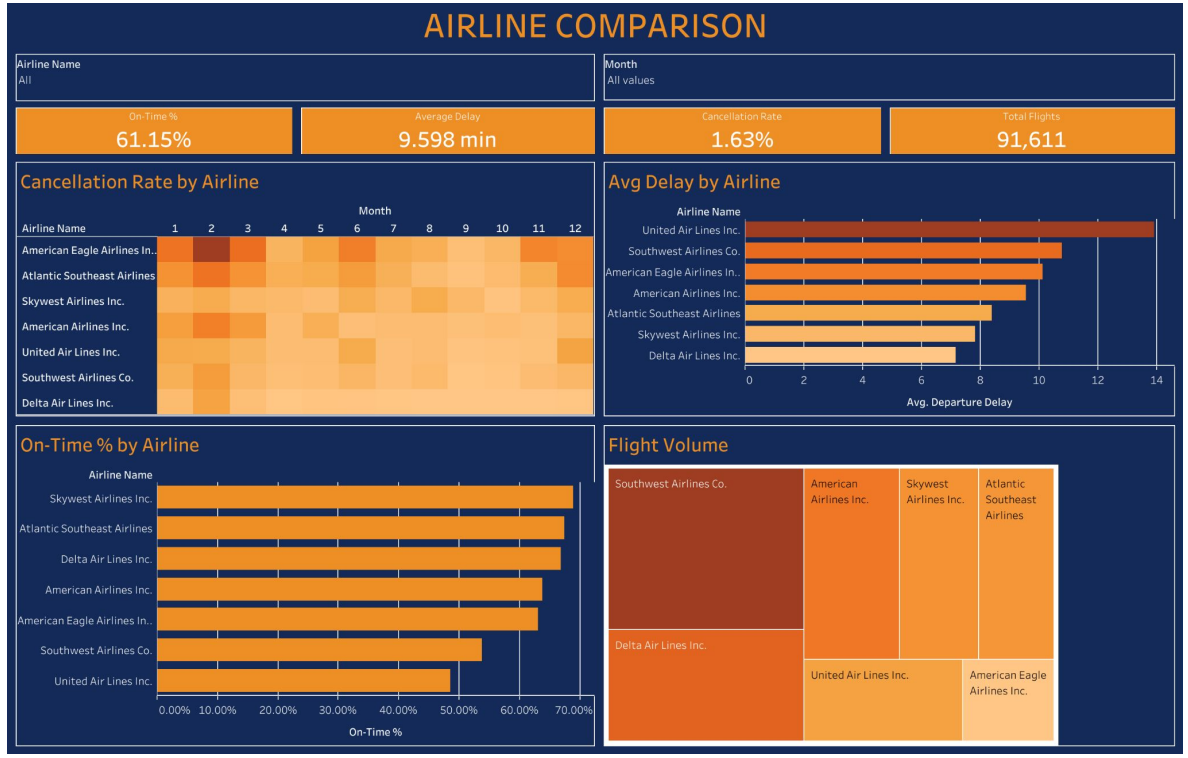


▸ **INTERACTIVE FILTERS AVAILABLE ON ALL 4 VIEWS**

Month range · Airline · Day-of-week · Time-of-day · Delay type · Airport · City

TAKEAWAY · Audience picks a filter → entire workbook re-renders in real time. Live on Tableau Public.

Two Views, One Decision-Ready Workbook



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Five Operational Levers — Each Tied to an Insight

Action → evidence behind it → expected impact. Numbered priorities the COO can act on this quarter.

		OTP / OPS	$ IMPACT
R1	**Enforce departure discipline at top-10 hubs** *Insight 01 — cascade r = 0.94*	+4–6 pp OTP	$8.4M avoided
R2	**Re-bias mission-critical flights to morning slots** *Insight 02 — 4× delay swing AM→PM*	↓ 8% missed connects	$1.6M avoided
R3	**Standardise turnaround SOPs at ORD / EWR / SFO** *Insight 03, 05 — ground ops dominate*	12–18% ↓ cascading delay	$4–6M avoided
R4	**Pre-position resources for known seasonal peaks** *Insight 02 — Jun–Aug & Dec–Feb peaks*	15% ↓ IRROPS recovery	$2M avoided
R5	**Trigger DOT review of high-cancellation carriers** *Test 2 — χ^2 confirms airline dependence*	Policy lever	Public-trust value

TAKEAWAY · Aggregate sample-level impact: ~$16M avoided cost + 4–6 pp OTP lift.

What This Unlocks — Sample, Operational, Network-Wide

Three layers of value, each with the math chain that produced it.

SAMPLE-LEVEL (100K)

$16 M

Avoided cost

Sum of R1–R4 dollar impacts.

NETWORK-WIDE EFFECT

+4–6 pp

OTP lift

Driven by R1 (departure discipline).

FULL BTS 2015 (5.8M FLIGHTS)

$900 M+

Network opportunity

Linear scale of sample impact ×58.

▸ HOW THE NUMBERS WORK (the math chain)

$74.20
FAA cost / delay-min

× 3 min
Reduction in dep delay (realistic at top-10 hubs)

× ~38K
Flights through top-10 hubs in sample

= $8.4M
Sample-level avoided cost from R1

× 58
Scale to full 5.8M-flight year

≈ $487M
From R1 alone — and R2-R4 push to $900M+

TAKEAWAY · Conservative assumption: 3-min reduction from on-stand discipline. Even at half that, the case clears.

Limitations, Next Steps & Team

Honest about what we cannot say from this data, and what would make it stronger.

✕ LIMITATIONS — what we can't say from this data

- ✕ Single-year (2015) — no post-pandemic dynamics
- ✕ 100K stratified sample — limited power for very-rare events
- ✕ 8% records use legacy FAA codes (no metadata join)
- ✕ Self-reported delay-cause buckets — known carrier bias
- ✕ No passenger-itinerary data — can't model knock-on miss-connects

✦ NEXT STEPS — what we'd build with more time

- ✦ Multi-year extension (2010–2024) — capture COVID shock
- ✦ ML predictor — gradient boosting on the OLS feature set
- ✦ Real-time SWIM-feed streaming dashboard for ops war-room
- ✦ Weather-data join (METAR/TAF) — disentangle true weather delay
- ✦ Causal inference layer for capex / SOP ROI estimation

👥 TEAM GROUP-19 · SECTION D

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Data Lead

M3
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Member 4
Analysis Lead

Member 5
Visualization Lead

Member 6
Strategy Lead

Member 7
PPT & Quality Lead

THANK YOU · QUESTIONS WELCOME

github.com/adamyatiwari12/SectionD_G19_FlightOps-Dashboard